INORGANIC CHEMISTRY CUMULATIVE EXAM
November 15, 1997

The most recent (11/10) general chemistry seminar featured a speaker of notable importance in the area of organometallic chemistry. The title of the talk was "New Forms of Coordinated Carbon: C_2-C_{20} Chains that Span Two Transition Metals".

1. What was the speaker's name and affiliation?

2. In most of the examples described during the talk, the metal centers were of the type $ReCp(NO)PR_3$. Several important features of this moiety were used to characterize the complexes and their properties. Describe how each of the following was used.
   a. NO IR bands
   b. P NMR
   c. Cyclic voltammetry

3. In addition to the above spectral handles, the speaker mentioned at least one other property of the metal centers that was advantageous to the work. Describe one.

4. Does this complex conform to the 18 electron rule when a carbon chain of the type described in the seminar is bonded to the metal center?

5. The speaker, on several occasions, compared the short chain complexes to polymer chains. What experimental criteria did he present to substantiate the claim that the longer chains resembled polymers.

6. What was the trend in stability of the complexes as a function of chain length and how did the speaker propose to alter the trend?

7. How did the speaker justify the work that he was doing?

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